

Scalable Low-Rank Tensor Completion

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Abstract

In this paper, we study low-rank tensor completion with partially observed entries. Inspired by the success of low-rank matrix completion and its tightest convex relaxation, we adapt it to low-rank tensor recovery. Different from matrix, singular value decomposition for tensor is way more sophisticated and computationally demanding. Therefore, we reformulate tensor nuclear norm into an upper-bounded surrogate function. The new objective exhibits favorable properties which admits the use of alternating minimization where in each update a closed optimal solution exists. Numerical experiment results and real-world examples are also provided to demonstrate the superiority of our proposed algorithm.

CCS Concepts

• Computing methodologies → Machine learning algorithms.

Keywords

tensor completion, optimization, CP decomposition, large-scale learning

ACM Reference Format:

Anonymous Author(s). 2026. Scalable Low-Rank Tensor Completion. In *Anonymous Submission*. ACM, New York, NY, USA, 12 pages.

1 Introduction

Many real-world data are intrinsically multiway, including medical images [28, 37], video and audio signals [13, 22], and spatiotemporal measurements [5, 7]. Tensors preserve interactions across these modes, and tensor completion seeks to recover their missing entries from a partially observed array [27, 32, 42]. A common assumption is that a small number of latent factors governs the data. This problem appears in data recovery [41], image and video completion [16, 35], and signal reconstruction [21]; tensor-valued inputs, parameters, and features are also pervasive in modern learning systems [1, 20]. Matrix completion is the order-two special case, however, extending low-rank completion from matrices to higher-order tensors is nontrivial and challenging because tensors have neither a unique rank nor a universal singular value decomposition. CP, Tucker, and tubal ranks capture different structures, while direct tensor nuclear-norm optimization is generally difficult. Penalizing nuclear norms of mode unfoldings is tractable but repeatedly computes SVDs of large matrices. Matricization can also obscure multiway structure, while the spectral operations and intermediate matrices can dominate runtime and memory for high-order or sparsely observed data. CP factorization avoids these SVDs and reduces storage and evaluates the loss only at observed entries, but standard CP least-squares models lack an explicit nuclear-norm connection. They also inherit factor-scaling indeterminacy: the same rank-one component can be represented by many reciprocal rescalings of its columns, complicating factor regularization.

Anonymous Submission,
2026.

In this paper, we develop an SVD-free CP completion framework whose factor regularizer uses every mode and upper-bounds the atomic tensor nuclear norm. It applies to arbitrary-order tensors without a tensor SVD or mode-wise SVDs, retaining a variational interpretation rather than introducing only an ad hoc factor penalty. For order greater than two, the bound need not be exact; we separate its gap into a *decomposition gap* and a *scaling gap*, characterize the best equivalent factor scaling, derive its expected gap under random component amplitudes, and recover the exact Burer–Monteiro representation for matrices.

The surrogate objective is jointly nonconvex but quadratic, smooth, and strongly convex in each factor block. A block can therefore be updated by solving a positive-definite linear system exactly or by a gradient step with a computable Lipschitz constant. This structure yields cyclic exact-minimization, gradient, and hybrid updates, together with monotonic descent, stationary-point convergence, and an $O(1/K)$ rate for the minimum squared stationarity measure. Our main contributions are summarized as follows.

- We introduce an arbitrary-order, SVD-free CP regularizer that is quadratic in each factor block and explicitly upper-bounds the atomic tensor nuclear norm.
- We characterize its decomposition and scaling gaps, including optimal factor balancing, the minimum scaling gap, and its expectation under random amplitudes; the matrix case recovers the exact variational form.
- Using computable Lipschitz bounds, we derive exact and gradient block updates and prove descent, convergence to stationary points, and an $O(1/K)$ rate for the squared stationarity measure.
- Experiments under extreme missingness and noise evaluate recovery accuracy, surrogate tightness, wall-clock convergence, and scaling with tensor dimensions, order, rank, and observations.

2 Related Work

Tensor rank models and convex surrogates. CP, Tucker, hierarchical Tucker, and tubal rank describe distinct low-dimensional structures [14, 26, 31, 40]: CP rank counts rank-one outer products, Tucker rank records mode-unfolding ranks, and t-SVD models promote low tubal rank. Consequently, tensor completion has no single analogue of matrix nuclear-norm minimization. Convex methods commonly penalize a weighted sum of mode-unfolding nuclear norms [17, 25] or a t-SVD tensor nuclear norm [40]. These models support recovery theory but require repeated SVDs of unfoldings or frequency-domain slices, which is a major computational and memory bottleneck for large, high-order, or sparsely observed tensors. Algorithms such as block-coordinate descent or ADMM distribute these spectral subproblems differently but do not eliminate them.

Factorization-based tensor completion. Factorized methods avoid large SVDs and can optimize only observed entries. CP-WOPT applies first-order optimization to an incomplete CP least-squares

model [2], while alternating methods solve one factor subproblem at a time by least squares. TMac simultaneously factorizes every mode unfolding and combines the resulting tensor estimates [34]. Although its mode-wise factorizations are parallelizable and SVD-free, TMac must maintain and initialize or adapt a factor pair and rank for each unfolding. Its storage, computation, and tuning burden therefore grow with tensor order. Its adaptive rank-adjustment strategy can reduce manual rank selection, but it does not remove these separate factorizations or their mode-dependent rank decisions. Moreover, TMac couples the mode-wise estimates through a shared completed tensor rather than through one shared CP factorization. It therefore does not directly exploit one set of cross-mode latent components or provide the atomic nuclear-norm interpretation pursued here. These tradeoffs make it a useful SVD-free unfolding baseline, but less attractive when order, memory, or per-dataset tuning is the primary constraint. Other alternatives include fixed-rank Tucker manifold optimization and tailored preconditioning [12], Bayesian CP priors that infer an effective rank and quantify uncertainty [23], and structure-aware proximal iterations that preserve sparse intermediates [36]. Scaled gradient methods offer local recovery guarantees under sampling, initialization, and incoherence assumptions [30]; structured regression subproblems can also be approximated by Richardson iteration and lifting [9]. Such methods avoid some spectral costs but can remain sensitive to rank, initialization, or factor conditioning.

Streaming and nonlinear completion. Online tensor-subspace methods update factors as slices arrive and impute without repeatedly solving a batch problem [19]. Neural models such as CoSTCo instead combine latent embeddings with convolutional components to learn nonlinear interactions in sparse tensors [15]. These directions target streaming data or greater expressiveness, whereas classical factorizations retain an explicit low-rank interpretation and permit direct convergence analysis.

Relation to our work. Motivated by the matrix nuclear-norm factorization [24, 33], our regularizer couples all CP factors and upper-bounds the atomic tensor nuclear norm. It avoids SVDs and retains a direct nuclear-norm connection absent from unregularized CP completion. For $\lambda > 0$, every block subproblem is smooth and strongly convex, supporting exact or gradient cyclic updates with monotonic-descent and stationary-point guarantees. Unlike independent unfolding factorizations, the same CP columns participate in every mode, so the model stores one coupled representation and exposes a single rank parameter. The price is joint nonconvexity and sensitivity to CP rank and initialization, which we control through per-dataset validation and report explicitly in the experiments.

3 Problem Formulation

3.1 Matrix Completion

We start with low-rank matrix completion with objective:

$$\min_{\mathbf{X}} \frac{1}{2} \|P_{\Omega}(\mathbf{X} - \mathbf{Y})\|_F^2 + \lambda \|\mathbf{X}\|_*, \quad (1)$$

where $\mathbf{Y}$ is the input data, $\mathbf{X}$ is to be recovered and P_{Ω} represents the projection onto observed subset. Here we use the nuclear norm ($\|\cdot\|_*$) as the surrogate of original rank function, due to its non-convexity and the nuclear norm is its tightest convex relaxation.

This objective function faces one challenge: because nuclear norm is the sum of all singular values, its direct calculation involves with SVD, which is generally considered very computationally expensive. In light of alleviating the cost, Burer-Monteiro form which uses the following theorem is introduced:

THEOREM 1. [24, 33] *The nuclear norm of a matrix $\|\mathbf{X}\|_*$ is equivalent to $\min_{\mathbf{A}\mathbf{B}^T=\mathbf{X}} \frac{1}{2} \|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2$ and the minimum is archived when $\mathbf{A} = \mathbf{U}\Sigma^{\frac{1}{2}}$ and $\mathbf{B} = \mathbf{V}\Sigma^{\frac{1}{2}}$, where $\mathbf{X} = \mathbf{U}\Sigma\mathbf{V}^T$ is the singular value decomposition of $\mathbf{X}$.*

Now with $\mathbf{X} = \mathbf{A}\mathbf{B}^T$, we can reformulate the problem as

$$F(\mathbf{A}, \mathbf{B}) = \frac{1}{2} \|P_{\Omega}(\mathbf{Y} - \mathbf{A}\mathbf{B}^T)\|_F^2 + \frac{\lambda}{2} (\|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2). \quad (2)$$

With the Burer-Monteiro form, we reach same solution as Eq. (1).

THEOREM 2. *Suppose there exist optimal solutions to the following two minimization problems:*

$$p = \min_{\mathbf{X} \in \mathbb{R}^{n_1 \times n_2}} \|\mathbf{X}\|_* + f(\mathbf{X}) \quad (3)$$

and

$$q = \min_{\mathbf{A} \in \mathbb{R}^{n_1 \times r}, \mathbf{B} \in \mathbb{R}^{n_2 \times r}} \frac{1}{2} (\|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2) + f(\mathbf{A}\mathbf{B}^T), \quad (4)$$

where $f : \mathbb{R}^{n_1 \times n_2} \rightarrow \mathbb{R}$, $r \geq \text{rank}(\mathbf{X}^*)$, $\mathbf{X}^*$ is an optimal solution to Problem (3). Then the above two minimization problems are equivalent in the sense that $p = q$.

We leave the proof to appendix. Given the above conclusion, for matrix completion problem, it is equivalent to optimize:

$$\min_{\mathbf{A}, \mathbf{B}} \frac{1}{2} \|P_{\Omega}(\mathbf{A}\mathbf{B}^T - \mathbf{Y})\|_F^2 + \frac{\lambda}{2} (\|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2), \quad (5)$$

which can be optimized alternatively as in each sub-problem, it boils down to a least square problem. The benefit of Burer-Monteiro fashion is apparent in terms of much less computation demand by avoiding singular value decomposition. After obtaining $\mathbf{A}^*, \mathbf{B}^*$, then $\mathbf{A}^*\mathbf{B}^{*T}$ will give the $\mathbf{X}^*$. By reformulating the objective function into Eq. (2), the key advantage of the reformulation is to avoid formidable *svd*, making the algorithm applicable to large-scale data.

3.2 Generalize to Tensor Completion

Now we generalize the completion problem to tensors. As a very natural generalization of Eq. (1), we can write the tensor completion problem with the objective function as

$$\min_{\mathbf{X}} \frac{1}{2} \|P_{\Omega}(\mathbf{X} - \mathbf{Y})\|_F^2 + \lambda \|\mathbf{X}\|_*, \quad (6)$$

with $\mathbf{X}$ being the restored tensor and $\mathbf{Y}$ the input data. Unfortunately, different from matrix, tensor nuclear norm is not uniquely defined and one possible way for tensor $\mathbf{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ is

$$\|\mathbf{X}\|_* = \{\inf \sum_{i=1}^R |\lambda_i| : \mathbf{X} = \sum_{i=1}^R \lambda_i \mathbf{u}_{1,i} \circ \dots \circ \mathbf{u}_{N,i}, \|\mathbf{u}_{k,i}\|_2 = 1\}, \quad (7)$$

where $\circ$ denotes outer product. The above definition indicates we can decompose the tensor into a linear combination of R rank-one tensors, and the minimal l_1 norm of the coefficients is defines as nuclear norm. This is coherent with the well known matrix nuclear norm and can be regarded as an extension. To see that, if we write

a matrix $\mathbf{X}$ of rank R as linear combination of norm-one rank-one terms, i.e. $\mathbf{X} = \sum_{i=1}^R \sigma_i \mathbf{W}_i$, where $\mathbf{W}_i$ satisfies $\mathbf{W}_i = \mathbf{u}_i \mathbf{v}_i^T$ and $\|\mathbf{u}\|_2 = \|\mathbf{v}\|_2 = 1$, we have

$$\begin{aligned} \|\mathbf{X}\|_* &= \left\| \sum_{i=1}^R \sigma_i \mathbf{W}_i \right\|_* \leq \sum_{i=1}^R \|\sigma_i \mathbf{W}_i\|_* \\ &\leq \sum_{i=1}^R \sigma_i \|\mathbf{W}_i\|_* = \sum_{i=1}^R \sigma_i \|\mathbf{u}_i \mathbf{v}_i^T\|_* = \sum_{i=1}^R \sigma_i. \end{aligned} \quad (8)$$

However, the tensor nuclear presents challenges. It is problematic to utilize this form in real-world applications due to the fact that computing the tensor nuclear norm is NP-hard [10].

Alternatively, another widely accepted tensor nuclear norm is defined as the sum of the nuclear norms of the tensor's all unfolding matrices along each mode. The most commonly used unfolding methods are CP decomposition [4, 29] and Tucker decomposition [18, 27]. In this paper, we use the CP method:

DEFINITION 1. [11] For a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$, its CP decomposition can be written as

$$\mathcal{X} = \sum_{r=1}^R \mathbf{a}_{1,r} \circ \mathbf{a}_{2,r} \circ \dots \circ \mathbf{a}_{N,r} = [[\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_N]], \quad (9)$$

where $\mathbf{a}_{i,r} \in \mathbb{R}^{I_i}$ is r -th column of $\mathbf{A}_i$ with $\mathbf{A}_i \in \mathbb{R}^{I_i \times R}$ and R is a positive integer. The n -th mode unfolding of $\mathcal{X}$ is

$$\mathbf{X}_{(n)} = \mathbf{A}_n (\mathbf{A}_N \odot \mathbf{A}_{N-1} \odot \dots \odot \mathbf{A}_{n+1} \odot \mathbf{A}_{n-1} \odot \dots \odot \mathbf{A}_1)^T, \quad (10)$$

where $\odot$ denotes the Khatri-Rao product.

Given the definition above, we use the following equation to depict the nuclear norm of low-rank tensor $\mathcal{X}$: $\|\mathcal{X}\|_* = \frac{1}{N} \sum_{i=1}^N \|\mathbf{X}_{(i)}\|_*$, whereas the objective function becomes

$$\min_{\mathcal{X}} \frac{1}{2} \|P_{\Omega}(\mathcal{X} - \mathcal{Y})\|_F^2 + \frac{1}{N} \sum_{i=1}^N \|\mathbf{X}_{(i)}\|_*. \quad (11)$$

We then apply Theorem 1 to the nuclear norm of each unfolding in Eq. (11). For a given i ,

$$\begin{aligned} \|\mathbf{X}_{(i)}\|_* &= \|\mathbf{A}_i (\mathbf{A}_N \odot \dots \odot \mathbf{A}_{i+1} \odot \mathbf{A}_{i-1} \odot \dots \odot \mathbf{A}_1)^T\|_* \\ &= \min_{\mathbf{A}_1, \dots, \mathbf{A}_N} \frac{1}{2} (\|\mathbf{A}_i\|_F^2 + \|\mathbf{A}_N \odot \dots \odot \mathbf{A}_{i+1} \odot \mathbf{A}_{i-1} \odot \dots \odot \mathbf{A}_1\|_F^2) \\ &= \min_{\mathbf{A}_1, \dots, \mathbf{A}_N} \frac{1}{2} (\|\mathbf{A}_i\|_F^2 + \text{tr}((\mathbf{A}_N^T \mathbf{A}_N) * \dots \\ &\quad * (\mathbf{A}_{i+1}^T \mathbf{A}_{i+1}) * (\mathbf{A}_{i-1}^T \mathbf{A}_{i-1}) * \dots * (\mathbf{A}_1^T \mathbf{A}_1))), \end{aligned} \quad (12)$$

where $*$ denotes the Hadamard product.

3.3 Two birds one stone: A unified formulation

We reformulate the low-rank tensor completion objective as

$$\begin{aligned} &\min_{\mathbf{A}_1, \dots, \mathbf{A}_N} F(\mathbf{A}_1, \dots, \mathbf{A}_N) \\ &= \min_{\mathbf{A}_1, \dots, \mathbf{A}_N} \frac{1}{2} \|P_{\Omega}(\mathcal{Y} - [[\mathbf{A}_1, \dots, \mathbf{A}_N]])\|_F^2 + \frac{\lambda}{2N} \sum_{i=1}^N (\|\mathbf{A}_i\|_F^2 \\ &\quad + \text{tr}((\mathbf{A}_N^T \mathbf{A}_N) * \dots * (\mathbf{A}_{i+1}^T \mathbf{A}_{i+1}) * (\mathbf{A}_{i-1}^T \mathbf{A}_{i-1}) * \dots * (\mathbf{A}_1^T \mathbf{A}_1))). \end{aligned} \quad (13)$$

The importance of (13) can be described from another perspective via the following proposition.

PROPOSITION 1. Suppose the nuclear rank of a tensor $\mathcal{X}$ is R , then for $\tilde{R} \geq R$, the minimum of the following optimization is a surrogate (upper bound) for $\|\mathcal{X}\|_*$:

$$\begin{aligned} &\min_{\{\mathbf{v}_{1,k}, \dots, \mathbf{v}_{N,k}\}_{k=1}^{\tilde{R}}} \frac{1}{2N} \sum_{k=1}^{\tilde{R}} \sum_{i=1}^N (\|\mathbf{v}_{i,k}\|_2^2 + \prod_{j \neq i} \|\mathbf{v}_{j,k}\|_2^2), \\ &\text{subject to } \mathcal{X} = \sum_{k=1}^{\tilde{R}} \mathbf{v}_{1,k} \circ \mathbf{v}_{2,k} \circ \dots \circ \mathbf{v}_{N,k}. \end{aligned} \quad (14)$$

PROOF. Suppose $\{\mathbf{v}_{1,k}, \dots, \mathbf{v}_{N,k}\}_{k=1}^{\tilde{R}}$ is an optimal solution to Problem (14). We define $u_{i,k} = \|\mathbf{v}_{i,k}\|_2$ and $\mu_k = \prod_{i=1}^N u_{i,k}$. Then for each $k \in \{1, \dots, \tilde{R}\}$ such that $\mu_k \neq 0$, let $\hat{\mathbf{v}}_{i,k} = \mathbf{v}_{i,k}/u_{i,k}$. We can now rewrite the tensor $\mathcal{X}$ as

$$\mathcal{X} = \sum_{k: \mu_k \neq 0} \mathbf{v}_{1,k} \circ \mathbf{v}_{2,k} \circ \dots \circ \mathbf{v}_{N,k} = \sum_{k: \mu_k \neq 0} \mu_k \hat{\mathbf{v}}_{1,k} \circ \hat{\mathbf{v}}_{2,k} \circ \dots \circ \hat{\mathbf{v}}_{N,k}. \quad (15)$$

By the definition of tensor nuclear norm in Eq. (7), Eq. (15) indicates that $\|\mathcal{X}\|_* \leq \sum_{k=1}^{\tilde{R}} \mu_k$. Therefore,

$$\begin{aligned} \|\mathcal{X}\|_* &\leq \sum_{k=1}^{\tilde{R}} \mu_k = \frac{1}{N} \sum_{k=1}^{\tilde{R}} N \prod_{i=1}^N u_{i,k} \leq \frac{1}{N} \sum_{k=1}^{\tilde{R}} \sum_{i=1}^N \left(\frac{u_{i,k}^2}{2} + \frac{\prod_{j \neq i} u_{j,k}^2}{2} \right) \\ &= \frac{1}{2N} \sum_{k=1}^{\tilde{R}} \sum_{i=1}^N (\|\mathbf{v}_{i,k}\|_2^2 + \prod_{j \neq i} \|\mathbf{v}_{j,k}\|_2^2) \\ &= \frac{1}{2N} \sum_{i=1}^N (\|\mathbf{A}_i\|_F^2 + \text{tr}((\mathbf{A}_N^T \mathbf{A}_N) * \dots * (\mathbf{A}_{i+1}^T \mathbf{A}_{i+1}) * (\mathbf{A}_{i-1}^T \mathbf{A}_{i-1}) * \dots * (\mathbf{A}_1^T \mathbf{A}_1))). \end{aligned} \quad (16)$$

□

To conclude, based on the two definitions of tensor nuclear norm, it merges to a unified reformulation to optimize the objective in Eq. (13) which enjoys several benign characteristics.

3.4 Tightness and Gap Analysis

We next characterize the gap between the tensor nuclear norm in (7) and the surrogate regularizer. This analysis separates the total approximation error into two components: the error associated with the chosen CP decomposition and the error induced by the scaling of the factor vectors.

Let $\mathcal{X} = \sum_{k=1}^{\tilde{R}} \mathbf{v}_{1,k} \circ \mathbf{v}_{2,k} \circ \dots \circ \mathbf{v}_{N,k}$, where $\mathbf{A}_i = [\mathbf{v}_{i,1}, \dots, \mathbf{v}_{i,\tilde{R}}]$, $u_{i,k} := \|\mathbf{v}_{i,k}\|_2$, $\mu_k := \prod_{i=1}^N u_{i,k}$. For every nonzero component, define the normalized vectors $\hat{\mathbf{v}}_{i,k} := \frac{\mathbf{v}_{i,k}}{u_{i,k}}$. Then

$$\mathcal{X} = \sum_{k=1}^{\tilde{R}} \mu_k \hat{\mathbf{v}}_{1,k} \circ \dots \circ \hat{\mathbf{v}}_{N,k}, \quad \|\hat{\mathbf{v}}_{i,k}\|_2 = 1.$$

Define the surrogate regularizer

$$\mathcal{R}(\mathbf{A}_1, \dots, \mathbf{A}_N) := \frac{1}{2N} \sum_{i=1}^N \left[\|\mathbf{A}_i\|_F^2 + \text{tr} \left(\prod_{j \neq i} \mathbf{A}_j^T \mathbf{A}_j \right) \right]. \quad (17)$$

The following result gives an exact decomposition of the gap.

PROPOSITION 2 (EXACT DECOMPOSITION OF THE SURROGATE GAP). For any CP factorization $\mathcal{X} = [[A_1, \dots, A_N]]$, the surrogate regularizer satisfies

$$\mathcal{R}(A_1, \dots, A_N) = \sum_{k=1}^{\tilde{R}} \mu_k + \frac{1}{2N} \sum_{k=1}^{\tilde{R}} \sum_{i=1}^N \left(u_{i,k} - \prod_{j \neq i} u_{j,k} \right)^2. \quad (18)$$

Consequently,

$$\mathcal{R}(A_1, \dots, A_N) - \|\mathcal{X}\|_* = \Delta_{\text{dec}} + \Delta_{\text{scale}}, \quad (19)$$

where

$$\Delta_{\text{dec}} := \sum_{k=1}^{\tilde{R}} \mu_k - \|\mathcal{X}\|_* \geq 0, \Delta_{\text{scale}} := \frac{1}{2N} \sum_{k=1}^{\tilde{R}} \sum_{i=1}^N \left(u_{i,k} - \prod_{j \neq i} u_{j,k} \right)^2 \geq 0. \quad (20)$$

We leave proof to appendix to save space. Proposition 2 shows that the total gap has two distinct sources. The decomposition gap Δ_{dec} measures the suboptimality of the chosen normalized CP representation relative to a nuclear-norm-minimizing representation. The scaling gap Δ_{scale} measures the additional looseness introduced by the quadratic inequality used to construct the surrogate.

3.4.1 Optimal scaling of each CP component. The CP representation is invariant to rescaling of its factor vectors. We next characterize the smallest possible surrogate value among all equivalent scalings of a fixed rank-one component.

Consider a component with coefficient $\mu > 0$, and let its factor norms satisfy $\prod_{i=1}^N u_i = \mu$. Its contribution to the surrogate is

$$g_N(u_1, \dots, u_N) = \frac{1}{2N} \sum_{i=1}^N \left(u_i^2 + \frac{\mu^2}{u_i^2} \right). \quad (21)$$

PROPOSITION 3 (MINIMUM GAP UNDER BALANCED SCALING). For a fixed coefficient $\mu > 0$,

$$\min_{\substack{u_i > 0 \\ \prod_{i=1}^N u_i = \mu}} g_N(u_1, \dots, u_N) = \phi_N(\mu), \text{ where } \phi_N(\mu) := \frac{1}{2} \left(\mu^{2/N} + \mu^{2-2/N} \right). \quad (22)$$

The minimum is attained at the balanced scaling

$$u_1 = \dots = u_N = \mu^{1/N}. \quad (23)$$

The corresponding minimum scaling gap is

$$g_N(\mu) := \phi_N(\mu) - \mu = \frac{1}{2} \left(\mu^{1/N} - \mu^{1-1/N} \right)^2. \quad (24)$$

We leave the proof to appendix.

The relative scaling gap can equivalently be written as

$$\frac{g_N(\mu)}{\mu} = \cosh \left(\left(1 - \frac{2}{N} \right) \log \mu \right) - 1. \quad (25)$$

For the matrix case $N = 2$, we have $g_2(\mu) = 0$ for every $\mu > 0$, which recovers the exact matrix Burer-Monteiro variational identity. For tensors of order $N > 2$, $g_N(\mu) = 0 \iff \mu = 1$. Therefore, even after optimally balancing the factor norms, the surrogate is generally not exact for higher-order tensors.

For coefficients close to one, a Taylor expansion gives

$$\frac{g_N(\mu)}{\mu} = \frac{1}{2} \left(1 - \frac{2}{N} \right)^2 (\log \mu)^2 + \mathcal{O}((\log \mu)^4). \quad (26)$$

Thus, the surrogate is tight when the logarithms of the CP coefficients are concentrated around zero.

3.4.2 Expected gap under random CP coefficients. We next provide an expectation analysis. Assume that the normalized CP representation attains the infimum in the definition of the tensor nuclear norm, so that $\|\mathcal{X}\|_* = \sum_{k=1}^{\tilde{R}} \mu_k$. Assume further that the CP coefficients $M_1, \dots, M_{\tilde{R}}$ are independent and identically distributed positive random variables, and that every rank-one component is represented using the balanced scaling in (23). Then

$$\mathbb{E}[\mathcal{R}(A_1, \dots, A_N) - \|\mathcal{X}\|_*] = \tilde{R} \mathbb{E}[g_N(M)]. \quad (27)$$

Using (24), we obtain

$$\mathbb{E}[g_N(M)] = \frac{1}{2} \mathbb{E}[M^{2/N}] + \frac{1}{2} \mathbb{E}[M^{2-2/N}] - \mathbb{E}[M], \quad (28)$$

provided that the corresponding moments are finite.

A convenient model for positive CP coefficients is the lognormal distribution. Suppose $\log M \sim \mathcal{N}(\eta, \sigma_\mu^2)$. Since $\mathbb{E}[M^p] = \exp \left(p\eta + \frac{p^2 \sigma_\mu^2}{2} \right)$, the expected gap per component is

$$\begin{aligned} \mathbb{E}[g_N(M)] &= \frac{1}{2} \exp \left[\frac{2\eta}{N} + \frac{1}{2} \left(\frac{2}{N} \right)^2 \sigma_\mu^2 \right] \\ &\quad + \frac{1}{2} \exp \left[\left(2 - \frac{2}{N} \right) \eta + \frac{1}{2} \left(2 - \frac{2}{N} \right)^2 \sigma_\mu^2 \right] - \exp \left(\eta + \frac{\sigma_\mu^2}{2} \right). \end{aligned}$$

In particular, when $\eta = 0$ and σ_μ^2 is small,

$$\mathbb{E}[g_N(M)] = \frac{1}{2} \left(1 - \frac{2}{N} \right)^2 \sigma_\mu^2 + o(\sigma_\mu^2). \quad (29)$$

Equation (29) shows that the expected gap increases with both the variability of the logarithmic CP coefficients and the tensor order.

4 Optimization and Convergence Analysis

$F(A_1, \dots, A_N)$ is not jointly convex for $A_1, \dots, A_N$. However, if we pick one certain A_i and fix the rest matrices $(A_{j \neq i})$, it is convex w.r.t. A_i . Research has indicated that, despite the objective function being nonconvex, it possesses a favorable geometric property where there are no spurious local minima [8]. In other words, a local minimum is also a global minimum. This allows us to utilize alternating minimization to optimize Eq. (13). For any i , we update the following objective:

$$\begin{aligned} F_i(A_i) &= \frac{1}{2} \|P_{\Omega_i}(Y_{(i)} - A_i(A_N \odot \dots \odot A_{i+1} \odot A_{i-1} \odot \dots \odot A_1)^T)\|_F^2 \\ &\quad + \frac{\lambda}{2N} (\|A_i\|_F^2 + \sum_{j \neq i} \text{tr}((h(i, j)) * (A_i^T A_j))), \end{aligned} \quad (30)$$

where P_{Ω_i} is the i^{th} unfolding of the observation mask, and $h(i, j)$ is a helper function that is defined as

$$\begin{aligned} h(i, j) &= (A_N^T A_N) * \dots * (A_{i+1}^T A_{i+1}) * (A_{i-1}^T A_{i-1}) * \dots \\ &\quad * (A_{j+1}^T A_{j+1}) * (A_{j-1}^T A_{j-1}) * \dots (A_1^T A_1). \end{aligned} \quad (31)$$

It can be reformulated into the vector form:

$$F_i(\mathbf{a}_i) = \frac{1}{2} \|P_{\Omega_{v,i}}(\text{vec}(\mathbf{Y}_{(i)}) - ((\mathbf{A}_N \odot \dots \odot \mathbf{A}_{i+1} \odot \mathbf{A}_{i-1} \odot \dots \odot \mathbf{A}_1) \otimes \mathbf{I}_{I_i}) \mathbf{a}_i\|_F^2 + \frac{\lambda}{2N} (\mathbf{a}_i^T \mathbf{a}_i + \mathbf{a}_i^T (\text{diag}(\text{diag}(\sum_{j \neq i} h(i, j))) \otimes \mathbf{I}_{I_i}) \mathbf{a}_i), \quad (32)$$

where $\mathbf{a}_i = \text{vec}(\mathbf{A}_i)$ and $P_{\Omega_{v,i}}$ is the vectorized i^{th} unfolding observation mask. We optimize the factor matrices cyclically using either an exact block-minimization step or a block-gradient step. The analysis below also permits a hybrid implementation in which different update types are used for different blocks.

For compactness, let $\mathcal{A} := (\mathbf{A}_1, \dots, \mathbf{A}_N)$, $\|\mathcal{A}\|^2 := \sum_{i=1}^N \|\mathbf{A}_i\|_F^2$, and define

$$\mathbf{B}_i := \mathbf{A}_N \odot \dots \odot \mathbf{A}_{i+1} \odot \mathbf{A}_{i-1} \odot \dots \odot \mathbf{A}_1. \quad (33)$$

Furthermore, let

$$\mathbf{y}_i := \text{vec}(\mathbf{Y}_{(i)}), \mathbf{P}_i := P_{\Omega_{v,i}}, \mathbf{K}_i := \mathbf{B}_i \otimes \mathbf{I}_{I_i}, \quad \mathbf{D}_i := \text{Diag}\left(\text{diag}\left(\sum_{j \neq i} h(i, j)\right)\right), \mathbf{M}_i := \mathbf{D}_i \otimes \mathbf{I}_{I_i}. \quad (34)$$

Here, $\text{diag}(\mathbf{G})$ extracts the diagonal of a matrix $\mathbf{G}$, whereas $\text{Diag}(\mathbf{x})$ forms a diagonal matrix from the vector $\mathbf{x}$.

For fixed $\mathcal{A}_{-i}$, the restriction of the global objective to the i^{th} block can be written as

$$F_i(\mathbf{a}_i; \mathcal{A}_{-i}) = \frac{1}{2} \|\mathbf{P}_i(\mathbf{y}_i - \mathbf{K}_i \mathbf{a}_i)\|_2^2 + \frac{\lambda}{2N} \mathbf{a}_i^T (\mathbf{I}_{I_R} + \mathbf{M}_i) \mathbf{a}_i + c_i(\mathcal{A}_{-i}), \quad (35)$$

where $c_i(\mathcal{A}_{-i})$ contains all terms independent of $\mathbf{a}_i$. Although this constant does not affect the block update, it is useful to retain it conceptually because $F_i(\mathbf{a}_i; \mathcal{A}_{-i}) = F(\mathbf{A}_1, \dots, \mathbf{A}_N)$.

Define

$$\mathbf{H}_i := \mathbf{K}_i^T \mathbf{P}_i^T \mathbf{P}_i \mathbf{K}_i + \frac{\lambda}{N} (\mathbf{I}_{I_R} + \mathbf{M}_i), \mathbf{b}_i := \mathbf{K}_i^T \mathbf{P}_i^T \mathbf{P}_i \mathbf{y}_i. \quad (36)$$

Then

$$\nabla_i F(\mathcal{A}) = \mathbf{H}_i \mathbf{a}_i - \mathbf{b}_i, \nabla_{ii}^2 F(\mathcal{A}) = \mathbf{H}_i. \quad (37)$$

When $\mathbf{P}_i$ is a symmetric idempotent masking matrix, $\mathbf{P}_i^T \mathbf{P}_i = \mathbf{P}_i$.

LEMMA 1 (BLOCK SMOOTHNESS AND STRONG CONVEXITY). Suppose $\lambda > 0$. For any fixed $\mathcal{A}_{-i}$, the block objective in (35) is strongly convex with modulus at least $\mu := \frac{\lambda}{N}$. Its exact block Lipschitz constant is $L_i^{\text{exact}} = \lambda_{\max}(\mathbf{H}_i)$. A convenient computable upper bound is

$$L_i^{\text{exact}} \leq \widehat{L}_i := \lambda_{\max}\left(\sum_{\ell \neq i} \mathbf{A}_\ell^T \mathbf{A}_\ell\right) + \frac{\lambda}{N} (1 + \|\mathbf{D}_i\|_2). \quad (38)$$

PROOF. Since $\mathbf{P}_i^T \mathbf{P}_i \succeq \mathbf{0}$ and $\mathbf{M}_i \succeq \mathbf{0}$, we have $\mathbf{H}_i \succeq \frac{\lambda}{N} \mathbf{I}_{I_R}$. Thus, the block objective is at least λ/N -strongly convex.

For the upper bound, first observe that

$$\lambda_{\max}(\mathbf{K}_i^T \mathbf{P}_i^T \mathbf{P}_i \mathbf{K}_i) = \|\mathbf{P}_i \mathbf{K}_i\|_2^2 \leq \|\mathbf{K}_i\|_2^2 = \|\mathbf{B}_i\|_2^2. \quad (39)$$

The Khatri–Rao identity gives $\mathbf{B}_i^T \mathbf{B}_i = \sum_{\ell \neq i} \mathbf{A}_\ell^T \mathbf{A}_\ell$, $\|\mathbf{B}_i\|_2^2 = \lambda_{\max}(\sum_{\ell \neq i} \mathbf{A}_\ell^T \mathbf{A}_\ell)$. Finally, $\|\mathbf{M}_i\|_2 = \|\mathbf{D}_i\|_2$, which proves (38). $\square$

Exact block minimization. Because $\mathbf{H}_i \succ \mathbf{0}$, the unique minimizer of the i^{th} block problem is characterized by $\mathbf{H}_i \mathbf{a}_i^+ = \mathbf{b}_i$. Equivalently, $\mathbf{a}_i^+ = \mathbf{H}_i^{-1} \mathbf{b}_i$. In implementation, it should be solved as a positive-definite linear system rather than by explicitly forming $\mathbf{H}_i^{-1}$.

Block-gradient update. Alternatively, given any valid upper bound $L_i \geq \lambda_{\max}(\mathbf{H}_i)$, one block-gradient step is

$$\mathbf{a}_i^+ = \mathbf{a}_i - \frac{1}{L_i} \nabla_i F(\mathcal{A}) = \mathbf{a}_i - \frac{1}{L_i} (\mathbf{H}_i \mathbf{a}_i - \mathbf{b}_i). \quad (40)$$

4.0.1 Cyclic update rule. Let $\mathcal{A}^k$ denote the factors at the beginning of the k^{th} sweep. Define the intermediate cyclic iterates

$$\mathcal{A}^{k,i} := (\mathbf{A}_1^{k+1}, \dots, \mathbf{A}_i^{k+1}, \mathbf{A}_{i+1}^k, \dots, \mathbf{A}_N^k), \mathcal{A}^{k,0} := \mathcal{A}^k, \mathcal{A}^{k,N} := \mathcal{A}^{k+1}. \quad (41)$$

Thus, when block i is updated, the latest available values of the preceding blocks are used.

Choosing EXACT for every block gives cyclic alternating minimization, or regularized alternating least squares. Choosing GRADIENT for every block gives cyclic block-gradient descent. Algorithm 1 also permits a hybrid strategy.

4.0.2 Convergence analysis. The following theorem establishes monotonic descent, convergence to a stationary point, and a sub-linear stationarity rate. It does not require the global objective to be convex or to have no spurious local minima.

THEOREM 3 (CONVERGENCE OF CYCLIC EXACT AND GRADIENT UPDATES). Suppose $\lambda > 0$ and every gradient update uses a valid constant $L_i^k \geq \lambda_{\max}(\mathbf{H}_i^k)$. Assume that the selected constants are uniformly bounded on the generated sequence: $L_i^k \leq \bar{L} < \infty$. This condition holds, for example, when (38) is used.

Let $\{\mathcal{A}^k\}$ be generated by Algorithm 1, where each block can use either an exact update or a gradient update. Define $F_{\inf} := \inf_{\mathcal{A}} F(\mathcal{A})$, $\mu := \frac{\lambda}{N}$. Then the following statements hold.

(1) The objective decreases monotonically, and

$$F(\mathcal{A}^k) - F(\mathcal{A}^{k+1}) \geq \frac{\mu}{2} \|\mathcal{A}^{k+1} - \mathcal{A}^k\|^2. \quad (42)$$

(2) The iterates are bounded, and

$$\sum_{k=0}^{\infty} \|\mathcal{A}^{k+1} - \mathcal{A}^k\|^2 \leq \frac{2}{\mu} (F(\mathcal{A}^0) - F_{\inf}), \|\mathcal{A}^{k+1} - \mathcal{A}^k\| \rightarrow 0. \quad (43)$$

(3) Every accumulation point of $\{\mathcal{A}^k\}$ is a stationary point of F .

(4) For every integer $K \geq 1$,

$$\min_{0 \leq k < K} \|\mathcal{A}^{k+1} - \mathcal{A}^k\|^2 \leq \frac{2(F(\mathcal{A}^0) - F_{\inf})}{\mu K}. \quad (44)$$

(5) Let Λ be a Lipschitz constant of the full gradient ∇F . Then

$$\min_{0 \leq k < K} \|\nabla F(\mathcal{A}^{k+1})\|^2 \leq \frac{4(\bar{L}^2 + N\Lambda^2)(F(\mathcal{A}^0) - F_{\inf})}{\mu K}. \quad (45)$$

Consequently, $\min_{0 \leq k < K} \|\nabla F(\mathcal{A}^{k+1})\| = \mathcal{O}(K^{-1/2})$.

(6) The complete sequence $\{\mathcal{A}^k\}$ converges to a stationary point.

We leave proof details in the appendix due to space limit.

5 Experiments

Our evaluation covers four complementary aspects: the optimization benefit of exact cyclic block minimization, recovery accuracy on large real tensors, time-accuracy behavior under matched compute budgets, and runtime and memory scaling with tensor size, order, rank, and number of observations. We refer to the resulting exact cyclic solver as *Cyclic Block Optimization* (CBO).

Algorithm 1 Cyclic Block Optimization for Tensor Completion

Require: Observed tensor $\mathbf{Y} \in \mathbb{R}^{I_1 \times \dots \times I_N}$, observation set Ω , target rank R , and regularization parameter $\lambda > 0$

Require: Block-update types $q_i \in \{\text{AM}, \text{GD}\}$ for $i \in [N]$

Ensure: Recovered tensor $\hat{\mathbf{X}}$

```

1: Initialize  $\mathcal{A}^0 = (\mathbf{A}_1^0, \dots, \mathbf{A}_N^0)$ 
2: Set  $k \leftarrow 0$ 
3: while the stopping criterion is not satisfied do
4:   Set  $\mathcal{A}^{k,0} \leftarrow \mathcal{A}^k$ 
5:   for  $i = 1, \dots, N$  do
6:     Construct the Khatri-Rao product
        $\mathbf{B}_i^k \leftarrow \mathbf{A}_N^{k,i-1} \odot \dots \odot \mathbf{A}_{i+1}^{k,i-1} \odot \mathbf{A}_{i-1}^{k,i-1} \odot \dots \odot \mathbf{A}_1^{k,i-1}$ 
7:     Set
        $\mathbf{K}_i^k \leftarrow \mathbf{B}_i^k \otimes \mathbf{I}_{I_i}$ 
8:     Compute
        $\mathbf{D}_i^k \leftarrow \text{Diag} \left( \text{diag} \left( \sum_{j \neq i} h^k(i, j) \right) \right)$ 
9:     Form
        $\mathbf{H}_i^k \leftarrow (\mathbf{K}_i^k)^\top \mathbf{P}_i^\top \mathbf{P}_i \mathbf{K}_i^k + \frac{\lambda}{N} (\mathbf{I}_{I_i R} + \mathbf{D}_i^k \otimes \mathbf{I}_{I_i})$ 
10:    Form
        $\mathbf{b}_i^k \leftarrow (\mathbf{K}_i^k)^\top \mathbf{P}_i^\top \mathbf{P}_i \text{vec}(\mathbf{Y}_{(i)})$ 
11:    if  $q_i = \text{AM}$  then
12:      Compute  $\mathbf{a}_i^{k+1}$  by solving
        $\mathbf{H}_i^k \mathbf{a}_i^{k+1} = \mathbf{b}_i^k$ 
13:    else
14:      Choose
        $L_i^k \geq \lambda_{\max}(\mathbf{H}_i^k)$ 
15:      Update
        $\mathbf{a}_i^{k+1} \leftarrow \mathbf{a}_i^k - \frac{1}{L_i^k} (\mathbf{H}_i^k \mathbf{a}_i^k - \mathbf{b}_i^k)$ 
16:    end if
17:    Reshape
        $\mathbf{A}_i^{k+1} \leftarrow \text{mat}_{I_i \times R}(\mathbf{a}_i^{k+1})$ 
18:    Set the cyclic intermediate iterate
        $\mathcal{A}^{k,i} \leftarrow (\mathbf{A}_1^{k+1}, \dots, \mathbf{A}_i^{k+1}, \mathbf{A}_{i+1}^k, \dots, \mathbf{A}_N^k)$ 
19:  end for
20:  Set  $\mathcal{A}^{k+1} \leftarrow \mathcal{A}^{k,N}$ 
21:  Set  $k \leftarrow k + 1$ 
22: end while
23: Set
        $\hat{\mathbf{X}} \leftarrow [[\mathbf{A}_1^k, \dots, \mathbf{A}_N^k]]$ 
24: return  $\hat{\mathbf{X}}$ 

```

5.1 Experimental Protocol

Data. Table 1 lists the five public tensors used in the main evaluation. Each tensor is independently min-max normalized to $[0, 1]$.



Figure 1: Top row: frames of grayscale video (3D) with massive missing observations; Bottom row: recovered frames.

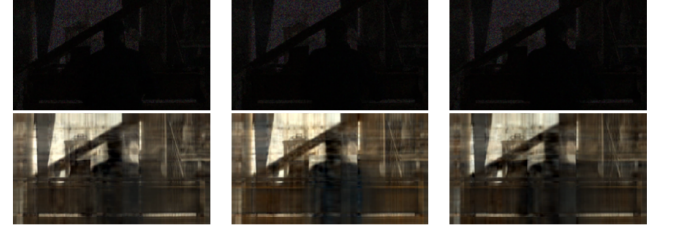


Figure 2: Top row: frames of colored video (4D) with massive missing observations; Bottom row: frames of video recovered.

Table 1: Real tensors in the main evaluation.

Tensor	Modality	Shape	Entries
CAVE Balloons	hyperspectral	256×256×31	2.03M
Xiph Bus	gray video	144×176×72	1.82M
Seattle	traffic	323×28×288	2.60M
Portland	traffic	1156×31×96	3.44M
Xiph City	color video	144×176×3×72	5.47M

The suite covers hyperspectral imaging, grayscale and color video, and two traffic systems. The fourth-order color video tests whether the implementation extends beyond order-three inputs.

Masks, noise, and metrics. For each tensor and seed, we sample an independent random-entry observation mask with either 20% or 10% observed entries, corresponding to 80% and 90% missingness. We use five seeds, shared by all methods. Measurement noise is added only to observed entries:

$$\mathbf{Y}_\omega = \mathbf{X}_\omega + \sigma(\max(\mathbf{X}) - \min(\mathbf{X}))\epsilon_\omega, \quad \epsilon_\omega \sim \mathcal{N}(0, 1), \quad (46)$$

where $\sigma \in \{0, 0.01, 0.05, 0.10\}$. Evaluation always uses the clean tensor on the missing entries $\mathcal{M} = \Omega^c$. The primary metric is

$$\text{RMSE}_{\mathcal{M}} = \sqrt{\frac{1}{|\mathcal{M}|} \left\| P_{\mathcal{M}}(\hat{\mathbf{X}} - \mathbf{X}) \right\|_F^2}. \quad (47)$$

Baselines and applicability. We compare with PALM [3] applied to the same factorized objective, CP-WOPT, HaLRTC [16], and LRTC-TNN [39]. TT-WOPT [38] is included for the fourth-order tensor, and LATC [6] is included only for traffic data. Mean filling

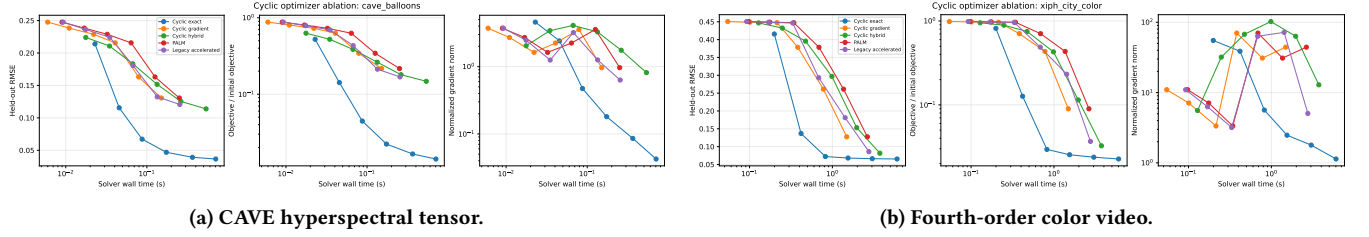


Figure 3: Held-out RMSE, normalized objective, and stationarity against solver wall time. Points are fresh deterministic runs at geometric sweep checkpoints; metric and stationarity computation are excluded from time.

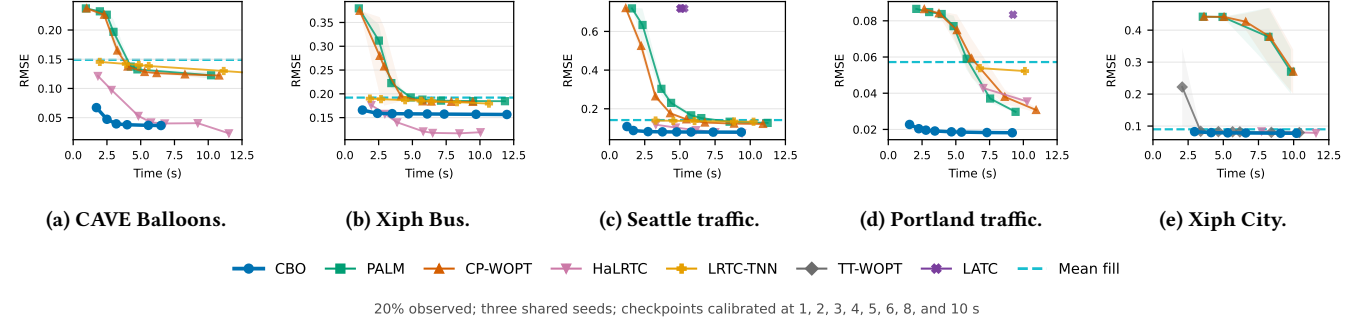


Figure 4: Missing-entry RMSE versus actual solver wall time at 80% missingness with clean observations. complete-sweep checkpoints calibrated at nominal 1, 2, 3, 4, 5, 6, 8, and 10 seconds; their horizontal positions are actual median solver times. The dashed horizontal line is mean filling. Lower is better.

is a non-iterative reference. We do not apply a modality-specific method outside its intended scope.

Hyperparameters and time matching. For each method–dataset pair, we reserve 10% of the observed entries for validation. Up to eight configurations are successively screened with 2-, 5-, and 10-second validation stages. Selection uses validation seed zero only; the selected configuration is then frozen and refit using all observed entries. CBO selects exact block minimization on all five tensors, with $(R, \lambda) = (8, 10^{-3})$ on CAVE and $(R, \lambda) = (4, 10^{-4})$ on the remaining tensors. Complete search spaces and selected baseline settings are provided in the supplement. Every iterative method receives a literal five-second solver budget. Data loading and metric computation are excluded from solver time. The experiment ran in Python 3.11 with NumPy 2.3 on a 10-core Apple M4 MacBook Air with 32GB RAM.

5.2 Extreme Missingness and Observation Noise

Table 2 gives the clean-observation results. At a five-second horizon, CBO has the lowest mean RMSE in nine of the ten dataset–missingness settings. The exception is Xiph Bus with 80% missing entries, where HaLRTC obtains 0.1402 RMSE compared with 0.1573 for CBO. At 90% missingness the ordering reverses on the same video. The largest margins occur on the traffic tensors; for example, on Portland with 80% missing entries, CBO obtains 0.0188 RMSE compared with 0.0507 for the strongest baseline.

Across all five datasets, two missingness levels, and four noise levels, CBO wins 36 of 40 dataset–condition cells and has mean rank 1.10. All four losses are the Xiph Bus conditions at 80% missingness.

Table 2: Missing-entry RMSE with clean observations. Values are mean $\pm$ standard deviation over five masks. “Other” is the strongest applicable iterative baseline in that row. All iterative methods receive the same five-second target.

Tensor	Missing	CBO	Other	Method
CAVE	80%	0.03737 $\pm$ 0.00055	0.05306 $\pm$ 0.00017	HaLRTC
CAVE	90%	0.03761 $\pm$ 0.00132	0.13482 $\pm$ 0.00028	HaLRTC
Bus	80%	0.15728 $\pm$ 0.00137	0.14022 $\pm$ 0.00019	HaLRTC
Bus	90%	0.15660 $\pm$ 0.00042	0.17958 $\pm$ 0.00025	HaLRTC
Seattle	80%	0.08013 $\pm$ 0.00138	0.10278 $\pm$ 0.00019	HaLRTC
Seattle	90%	0.07927 $\pm$ 0.00250	0.12717 $\pm$ 0.00009	HaLRTC
Portland	80%	0.01878 $\pm$ 0.00043	0.05069 $\pm$ 0.00003	HaLRTC
Portland	90%	0.01913 $\pm$ 0.00117	0.05360 $\pm$ 0.00002	HaLRTC
City	80%	0.08024 $\pm$ 0.00149	0.08209 $\pm$ 0.00048	TT-WOPT
City	90%	0.07895 $\pm$ 0.00093	0.08181 $\pm$ 0.00077	TT-WOPT

HaLRTC wins those four cells and has mean rank 2.10. TT-WOPT has mean rank 2.00 on its eight applicable fourth-order cells. The calibrated CBO checkpoint lies in the predeclared 0.8–1.25 timing window in all 40 conditions. At $\sigma = 0.10$, the median RMSE increase of CBO relative to clean observations is only 0.44%, although the mean is 6.87% because Portland is more noise-sensitive. The increase reaches 19.85% and 30.72% on Portland at 80% and 90% missingness, respectively; its absolute RMSE nevertheless remains lower than all baselines. Figure 5 shows the complete rank matrix.

The apparently small change from 80% to 90% missingness for several factorized methods should be interpreted as a wall-time result, not an iteration-matched result: fewer observations make each sweep cheaper, allowing more sweeps within five seconds.

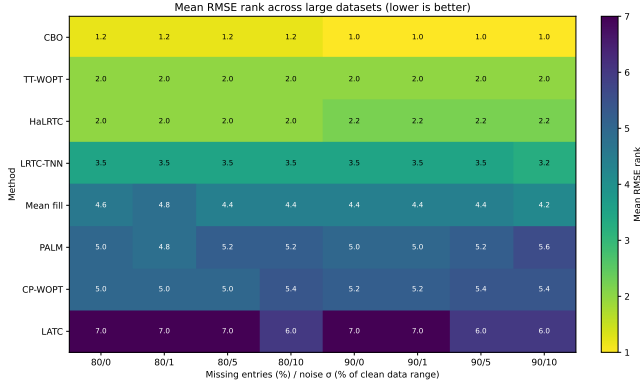


Figure 5: Mean RMSE rank across five datasets. Columns denote missing percentage/noise standard deviation as a percentage of the normalized data range. Lower is better. Dataset-specific methods are averaged only over applicable tensors.

Table 3: Optimizer ablation at 32 sweeps. Each cell reports missing-entry RMSE / median solver seconds; RMSE is mean over five shared masks.

Optimizer	CAVE	City
Exact cyclic	0.0366/0.706	0.0654/6.025
Gradient cyclic	0.1306/0.151	0.1274/1.490
Hybrid cyclic	0.1138/0.537	0.0816/3.739
PALM	0.1306/0.251	0.1274/2.641
Previous accelerated	0.1207/0.255	0.0859/2.755

5.3 Cyclic Update Ablation

We isolate the optimizer using CAVE Balloons ($64 \times 64 \times 31$) and Xiph City ($64 \times 80 \times 3 \times 48$), both with 90% missing entries. Every variant receives the same observations, rank $R = 8$, $\lambda = 10^{-3}$, and initial factors. We compare exact cyclic updates, Lipschitz gradient cyclic updates, an alternating exact/gradient hybrid, PALM, and the previous accelerated update. Fresh deterministic fits are evaluated after 1, 2, 4, 8, 16, and 32 complete sweeps for each of five seeds.

Exact updates cost more per sweep but reach substantially lower recovery error and lower stationarity measures by the final checkpoint. On CAVE, their normalized gradient norm is 4.24×10^{-2} , compared with 6.32×10^{-1} for the previous accelerated update and 9.64×10^{-1} for gradient cyclic/PALM. On City, exact cyclic reaches 1.14, compared with 5.05 for the previous update and 44.3 for gradient cyclic/PALM. Across all 300 fitted trajectories, no objective increase exceeds the numerical tolerance, empirically confirming the monotonic-descent behavior used in our analysis.

5.4 Accuracy and Time

The five-second comparison in Section 5.2 summarizes one operating point. To examine how recovery evolves with additional computation, we reuse the clean-observation, 80%-missing trajectories from the tuned large-tensor experiment. Each method-dataset configuration is selected before testing and then frozen. We evaluate fresh deterministic fits on three shared masks at nominal 1-, 2-,

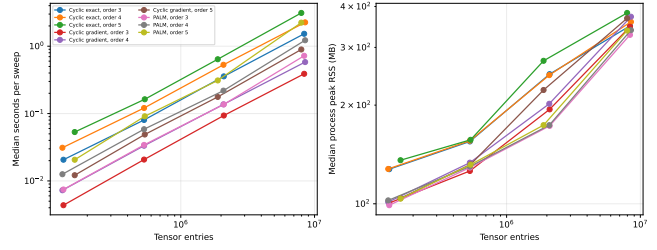


Figure 6: Median seconds per sweep and fresh-process peak resident memory versus tensor entries. Lines distinguish optimizer and tensor order. Both axes are logarithmic.

3-, 4-, 5-, 6-, 8-, and 10-second horizons. The horizontal coordinate in Figure 4 is the median measured solver time. If adjacent targets map to the same complete-sweep count, that fresh fit appears only once. Across the 38 dataset-budget cells containing at least one iterative method whose median solver time lies in the predeclared timing window, CBO obtains the lowest mean RMSE in 26. HaLRTC wins nine cells, while PALM, CP-WOPT, and TT-WOPT each win one. No method falls inside the timing window for the one-second Portland and Xiph City cells, which are excluded from this count.

5.5 Scalability

We generate sparse observations directly from CP factors, without materializing a dense tensor. We vary tensor order in $\{3, 4, 5\}$ and total entries from approximately 1.3×10^5 to 8.5×10^6 at rank 8 and 10% observation rate. Separate sweeps vary rank $R \in \{4, 8, 16, 32\}$ and observation rate $p \in \{0.05, 0.10, 0.20, 0.40\}$ on an order-three tensor with 2.10 million entries. Every case uses three seeds and three complete sweeps. Each method-case-seed combination runs in a fresh single-thread process so peak resident memory is not inherited from previous fits. Figure 6 shows nearly linear time growth in the number of tensor entries. The empirical time exponents for exact cyclic updates are 1.043, 1.031, and 1.049 for orders 3, 4, and 5, respectively. At the largest tested sizes, exact cyclic updates require 1.53, 2.26, and 3.12 seconds per sweep, with peak resident memory of 349, 359, and 382 MB. The corresponding time exponents are 1.047–1.094 for cyclic gradient and 1.084–1.182 for PALM. Rank is the principal cost driver for exact block solves. At 2.10 million entries and 10% observations, increasing R from 4 to 32 increases exact-cyclic time from 0.211 to 8.689 seconds per sweep and peak memory from 165 to 1162 MB. Increasing the observation rate from 5% to 40% at rank 8 increases time from 0.249 to 2.730 seconds per sweep. All 180 scaling fits are numerically monotone.

6 Conclusion

In this paper, we study tensor completion problem by making use of low-rank property which is widely assumed intrinsically such as in spatiotemporal dataset and media stream data. However, due to its high computation cost obtaining singular value, we propose a novel reformulation as a surrogate function, which enjoys benign properties. Besides, we propose alternating minimization approach with acceleration for optimization with provable theoretical guarantee. Experiments on various real-world data demonstrate its huge potential in dealing with large-scale multidimensional data.

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7 Proof of Theorem 2

PROOF. We only consider the simple case where $r = \text{rank}(\mathbf{X}^*) = \min(n_1, n_2)$. Other cases can be easily proved by minor modifications of this proof.

For an optimal solution $\mathbf{X}^*$ to Problem (3), suppose $\mathbf{X}^* = \mathbf{U}\Sigma\mathbf{V}^T$ is an SVD of $\mathbf{X}$. Choose $\mathbf{A}' = \mathbf{U}\Sigma^{\frac{1}{2}}, \mathbf{B}' = \mathbf{V}\Sigma^{\frac{1}{2}}$, we have

$$q \leq \frac{1}{2}(\|\mathbf{A}'\|_F^2 + \|\mathbf{B}'\|_F^2) + f(\mathbf{A}'\mathbf{B}'^T) = \|\mathbf{X}^*\|_* + f(\mathbf{X}^*) = p. \quad (48)$$

Similarly, suppose $\mathbf{A}^*, \mathbf{B}^*$ is an optimal solution to Problem (4), i.e., $q = \frac{1}{2}(\|\mathbf{A}^*\|_F^2 + \|\mathbf{B}^*\|_F^2) + f(\mathbf{A}^*\mathbf{B}^{*T})$, and pick $\mathbf{X}' = \mathbf{A}^*\mathbf{B}^{*T}$. Since $\|\mathbf{X}'\|_* \leq \frac{1}{2}(\|\mathbf{A}^*\|_F^2 + \|\mathbf{B}^*\|_F^2)$ by Theorem 1, we have

$$p \leq \|\mathbf{X}'\|_* + f(\mathbf{X}') \leq \frac{1}{2}(\|\mathbf{A}^*\|_F^2 + \|\mathbf{B}^*\|_F^2) + f(\mathbf{A}^*\mathbf{B}^{*T}) = q. \quad (49)$$

Since both $p \leq q$ and $q \leq p$ hold, we have $p = q$. $\square$

8 Proof of Proposition 2

PROOF. Using $\|A_i\|_F^2 = \sum_{k=1}^{\tilde{R}} u_{i,k}^2$ and the identity $\text{tr}(*_{j \neq i} A_j^\top A_j) = \sum_{k=1}^{\tilde{R}} \prod_{j \neq i} u_{j,k}^2$, we obtain

$$\mathcal{R}(A_1, \dots, A_N) = \frac{1}{2N} \sum_{k=1}^{\tilde{R}} \sum_{i=1}^N \left(u_{i,k}^2 + \prod_{j \neq i} u_{j,k}^2 \right). \quad (50)$$

For each component k ,

$$\frac{1}{2N} \sum_{i=1}^N \left(u_{i,k}^2 + \prod_{j \neq i} u_{j,k}^2 \right) - \mu_k = \frac{1}{2N} \sum_{i=1}^N \left[u_{i,k}^2 + \left(\prod_{j \neq i} u_{j,k} \right)^2 - 2u_{i,k} \prod_{j \neq i} u_{j,k} \right] = \frac{1}{2N} \sum_{i=1}^N \left(u_{i,k} - \prod_{j \neq i} u_{j,k} \right)^2, \quad (51)$$

where we used $u_{i,k} \prod_{j \neq i} u_{j,k} = \prod_{j=1}^N u_{j,k} = \mu_k$. Summing (51) over k proves (18).

Moreover, the normalized CP representation $\mathcal{X} = \sum_{k=1}^{\tilde{R}} \mu_k \widehat{v}_{1,k} \circ \dots \circ \widehat{v}_{N,k}$ is feasible in the definition of the tensor nuclear norm. Hence, $\|\mathcal{X}\|_* \leq \sum_{k=1}^{\tilde{R}} \mu_k$. Subtracting $\|\mathcal{X}\|_*$ from (18) gives (19). $\square$

9 Proof of Proposition 3

PROOF. Let $x_i = u_i^2$. Since $\prod_{i=1}^N x_i = \mu^2$, the arithmetic–geometric mean inequality gives

$$\frac{1}{N} \sum_{i=1}^N x_i \geq \left(\prod_{i=1}^N x_i \right)^{1/N} = \mu^{2/N}.$$

Similarly,

$$\frac{1}{N} \sum_{i=1}^N \frac{\mu^2}{x_i} \geq \left(\prod_{i=1}^N \frac{\mu^2}{x_i} \right)^{1/N} = \mu^{2-2/N}.$$

Substituting these inequalities into (21) gives

$$q_N(u_1, \dots, u_N) \geq \frac{1}{2} \left(\mu^{2/N} + \mu^{2-2/N} \right).$$

Equality in both arithmetic–geometric mean inequalities holds if and only if

$$u_1^2 = \dots = u_N^2 = \mu^{2/N},$$

which proves (23) and (22). Finally,

$$\phi_N(\mu) - \mu = \frac{1}{2} \left(\mu^{2/N} + \mu^{2-2/N} - 2\mu \right) = \frac{1}{2} \left(\mu^{1/N} - \mu^{1-1/N} \right)^2.$$

$\square$

10 More on Gap Analysis

10.0.1 Expected objective gap under random observations. Let

$$J_*(\mathcal{X}; \Omega) = \frac{1}{2} \|P_\Omega(\mathcal{Y} - \mathcal{X})\|_F^2 + \lambda \|\mathcal{X}\|_* \quad (52)$$

denote the original nuclear-norm-regularized objective, and let

$$J_{\mathcal{R}}(A_1, \dots, A_N; \Omega) = \frac{1}{2} \|P_\Omega(\mathcal{Y} - [[A_1, \dots, A_N]])\|_F^2 + \lambda \mathcal{R}(A_1, \dots, A_N). \quad (53)$$

For the same represented tensor $\mathcal{X} = [[A_1, \dots, A_N]]$, the data-fitting terms are identical. Therefore, for every observation set Ω ,

$$J_{\mathcal{R}}(A_1, \dots, A_N; \Omega) - J_*(\mathcal{X}; \Omega) = \lambda [\mathcal{R}(A_1, \dots, A_N) - \|\mathcal{X}\|_*]. \quad (54)$$

If every tensor entry is observed independently with probability p , then

$$\mathbb{E}_\Omega [\|P_\Omega(\mathcal{Y} - \mathcal{X})\|_F^2] = p \|\mathcal{Y} - \mathcal{X}\|_F^2.$$

Consequently,

$$\mathbb{E}_\Omega [J_{\mathcal{R}}] = \frac{p}{2} \|\mathcal{Y} - \mathcal{X}\|_F^2 + \lambda \mathcal{R}(A_1, \dots, A_N), \quad (55)$$

$$\mathbb{E}_\Omega [J_*] = \frac{p}{2} \|\mathcal{Y} - \mathcal{X}\|_F^2 + \lambda \|\mathcal{X}\|_*, \quad (56)$$

and hence

$$\mathbb{E}_\Omega [J_{\mathcal{R}} - J_*] = \lambda [\mathcal{R}(A_1, \dots, A_N) - \|\mathcal{X}\|_*]. \quad (57)$$

Thus, random sampling affects the expected data-fitting term, but does not change the regularization gap for a fixed represented tensor.

10.0.2 A computable interval for the unknown gap. Although the tensor nuclear norm is generally difficult to evaluate, the gap can be bounded using quantities obtained directly from the learned factors. Define $U_{\text{CP}} := \sum_{k=1}^R \mu_k$. Then

$$\|\mathcal{X}\|_* \leq U_{\text{CP}} \leq \mathcal{R}(A_1, \dots, A_N).$$

Moreover, every matrix unfolding provides a lower bound:

$$\max_{1 \leq i \leq N} \|\mathcal{X}_{(i)}\|_* \leq \|\mathcal{X}\|_*.$$

Therefore,

$$\mathcal{R}(A_1, \dots, A_N) - U_{\text{CP}} \leq \mathcal{R}(A_1, \dots, A_N) - \|\mathcal{X}\|_* \leq \mathcal{R}(A_1, \dots, A_N) - \max_{1 \leq i \leq N} \|\mathcal{X}_{(i)}\|_*. \quad (58)$$

The lower endpoint is precisely the scaling gap Δ_{scale} , whereas the upper endpoint can be evaluated by computing the nuclear norms of the unfolding matrices. This interval can therefore be used to quantify the empirical tightness of the surrogate in the experiments.

11 Proof of Theorem 3

PROOF. We first establish sufficient decrease for one block update. Let $\Delta_i^k := \mathbf{a}_i^{k+1} - \mathbf{a}_i^k$.

For an exact update, the block objective is μ -strongly convex and $\mathbf{a}_i^{k+1}$ is its unique minimizer. Hence,

$$F(\mathcal{A}^{k,i-1}) - F(\mathcal{A}^{k,i}) \geq \frac{\mu}{2} \|\mathbf{a}_i^{k+1} - \mathbf{a}_i^k\|_2^2 = \frac{\mu}{2} \|\Delta_i^k\|_2^2. \quad (59)$$

For a gradient update, block smoothness and $L_i^k \geq \lambda_{\max}(\mathbf{H}_i^k)$ give

$$F(\mathcal{A}^{k,i}) \leq F(\mathcal{A}^{k,i-1}) + \langle \nabla_i F(\mathcal{A}^{k,i-1}), \Delta_i^k \rangle + \frac{L_i^k}{2} \|\Delta_i^k\|_2^2.$$

Since $\Delta_i^k = -\frac{1}{L_i^k} \nabla_i F(\mathcal{A}^{k,i-1})$, we obtain

$$F(\mathcal{A}^{k,i-1}) - F(\mathcal{A}^{k,i}) \geq \frac{1}{2L_i^k} \left\| \nabla_i F(\mathcal{A}^{k,i-1}) \right\|_2^2 = \frac{L_i^k}{2} \|\Delta_i^k\|_2^2. \quad (60)$$

Because $L_i^k \geq \lambda_{\max}(\mathbf{H}_i^k) \geq \lambda_{\min}(\mathbf{H}_i^k) \geq \mu$, we have

$$F(\mathcal{A}^{k,i-1}) - F(\mathcal{A}^{k,i}) \geq \frac{\mu}{2} \|\Delta_i^k\|_2^2. \quad (61)$$

Thus, (61) holds for both update types. Summing over $i = 1, \dots, N$ yields

$$F(\mathcal{A}^k) - F(\mathcal{A}^{k+1}) = \sum_{i=1}^N [F(\mathcal{A}^{k,i-1}) - F(\mathcal{A}^{k,i})] \geq \frac{\mu}{2} \sum_{i=1}^N \|\Delta_i^k\|_2^2 = \frac{\mu}{2} \|\mathcal{A}^{k+1} - \mathcal{A}^k\|^2, \quad (62)$$

which proves (42).

The objective is bounded below. Moreover, all regularization terms are nonnegative, and therefore

$$F(\mathcal{A}) \geq \frac{\lambda}{2N} \sum_{i=1}^N \|\mathbf{A}_i\|_F^2. \quad (63)$$

Thus, the initial level set

$$\mathcal{L}_0 := \{\mathcal{A} : F(\mathcal{A}) \leq F(\mathcal{A}^0)\}$$

is compact. Since the objective decreases after every block update, all iterates and intermediate iterates remain in $\mathcal{L}_0$.

Summing (42) from $k = 0$ to infinity gives (43). Equation (44) follows by averaging over the first K sweeps.

It remains to relate the step norm to stationarity. For each block, define a reference point $\mathcal{Z}_i^k$ as follows:

$$\mathcal{Z}_i^k = \begin{cases} \mathcal{A}^{k,i}, & \text{if block } i \text{ is updated exactly,} \\ \mathcal{A}^{k,i-1}, & \text{if block } i \text{ uses a gradient update.} \end{cases}$$

For an exact update,

$$\nabla_i F(\mathcal{Z}_i^k) = \mathbf{0}.$$

For a gradient update,

$$\nabla_i F(\mathcal{Z}_i^k) = -L_i^k \Delta_i^k.$$

Consequently, in both cases,

$$\|\nabla_i F(\mathcal{Z}_i^k)\|_2 \leq \bar{L} \|\Delta_i^k\|_2. \quad (64)$$

Because F is continuously differentiable and all intermediate iterates lie in a compact set, its gradient is Lipschitz on a compact set containing their convex hull. Let Λ be a corresponding Lipschitz constant. Since

$$\|\mathcal{A}^{k+1} - \mathcal{Z}_i^k\| \leq \|\mathcal{A}^{k+1} - \mathcal{A}^k\|,$$

we have

$$\|\nabla_i F(\mathcal{A}^{k+1})\|_2 \leq \|\nabla_i F(\mathcal{Z}_i^k)\|_2 + \Lambda \|\mathcal{A}^{k+1} - \mathcal{Z}_i^k\| \leq \bar{L} \|\Delta_i^k\|_2 + \Lambda \|\mathcal{A}^{k+1} - \mathcal{A}^k\|. \quad (65)$$

Using $(a + b)^2 \leq 2a^2 + 2b^2$ and summing over all blocks gives

$$\|\nabla F(\mathcal{A}^{k+1})\|^2 \leq 2 \left(\bar{L}^2 + N\Lambda^2 \right) \|\mathcal{A}^{k+1} - \mathcal{A}^k\|^2. \quad (66)$$

Combining (66) with (44) proves (45). It also shows that

$$\|\nabla F(\mathcal{A}^{k+1})\| \longrightarrow 0.$$

Because the sequence is bounded, every accumulation point is therefore stationary.

Finally, F is a polynomial in the entries of the factor matrices and is thus a Kurdyka–Łojasiewicz function. The sufficient-decrease condition (42), the relative-error condition (66), and boundedness imply that the sequence has finite length and converges to a single stationary point. $\square$

For the pure block-gradient variant, a slightly sharper intermediate stationarity bound can be stated. Define

$$\mathcal{G}_k^2 := \sum_{i=1}^N \left\| \nabla_i F(\mathcal{A}^{k,i-1}) \right\|_2^2. \quad (67)$$

COROLLARY 1 (RATE FOR CYCLIC BLOCK-GRADIENT DESCENT). *Suppose every block uses the gradient update (40) and $L_i^k \leq \bar{L}$. Then*

$$\min_{0 \leq k < K} \mathcal{G}_k^2 \leq \frac{2\bar{L} (F(\mathcal{A}^0) - F_{\inf})}{K}. \quad (68)$$

PROOF. From (60),

$$F(\mathcal{A}^k) - F(\mathcal{A}^{k+1}) \geq \sum_{i=1}^N \frac{1}{2L_i^k} \left\| \nabla_i F(\mathcal{A}^{k,i-1}) \right\|_2^2 \geq \frac{1}{2\bar{L}} \mathcal{G}_k^2. \quad (69)$$

Summing over $k = 0, \dots, K - 1$ and taking the minimum proves the result. $\square$